INCH-POUND
MIL-PRF-62411/2A(AT)
7 July 1997
SUPERSEDING
MIL-H-62411/2(AT)
5 July 1985

PERFORMANCE SPECIFICATION SHEET

HEAT EXCHANGER, NBC SYSTEM
CONDENSER P/N M62411/2
(ENGINEERING P/N 12336772)

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-PRF-62411A(AT) listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS

Design and construction. The heat exchanger shall conform to drawing hierarchy of which Drawing 12336772 is the top drawing.

Weight. The weight of the heat exchanger shall not exceed 6 pounds (lb) (2.72 kilograms (kg)) (see 4.6.3).

Performance.

Leakage.

External leakage. The pneumatic pressure applied to the hot air passages of the heat exchanger unit shall be 39 pounds per square inch gage (psig) (268.9 kilopascals (kPa)) (see 4.6.4.1).

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Internal leakage. The pressure applied to the hot air passages shall be 39 psig. The pressure loss shall not exceed 20 psig (137.9 kPa) in the first 20 seconds (see 4.6.4.2).

Proof pressure. The proof pressure shall be 59 psig (406.8 kPa)) (see 4.6.5).

Burst pressure. The burst pressure shall be 120 psig (827.4 kPa) at 126 degrees Fahrenheit (°F) (52 degrees Celsius (°C)) (see 4.6.6).

Heat transfer. Heat transfer performance shall be not less than that shown in figure 1 (see 4.6.7).

Air pressure loss.

Hot air. The hot air pressure loss shall not exceed that shown in figure 2 (see 4.6.8).

Cold air. The cold air pressure loss shall not exceed that shown in figure 2A (see 4.6.8).

Leakage.

External leakage. A pneumatic pressure of 39 to 40 psig (268.9 to 275.8 kPa) shall be applied to the hot air passages.

Internal leakage. A pneumatic pressure of 39 to 40 psig shall be applied to the hot air passages.

Proof pressure. The test pressure shall be 59 to 62 psig (406.8 to 427.5 kPa).

Burst pressure. The oven temperature shall be 126°F (52°C). The test pressure shall be 120 to 124 psig (827.4 to 854.9 kPa).

Heat transfer. To determine conformance to 3.5.4, the condenser shall operate at each combination of the steady state conditions specified in table VI. The measured hot air outlet temperature shall be equal to or greater than that specified in table VI.

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TABLE VI. Heat transfer.

Initial air temperature °F (°C)		Air flow lb/min (kg/min)		Hot air outlet outlet temperature
Hot	Cold	Hot	Cold	°F ±10°F (°C ±5.6°C)
136 (58)		10 (4.5)	15 (6.8)	66 (19)
136			30 (13.6)	76 (24)
136			52.2 (23.7)	86 (30)
136	Lab	14.7 (6.7)	15	50 (10)
136	ambient		30	60 (16)
136			52.2	70 (21)
136		20 (90.7)	15	40 (4)
136			30	48 (9)
136			52.2	57 (14)

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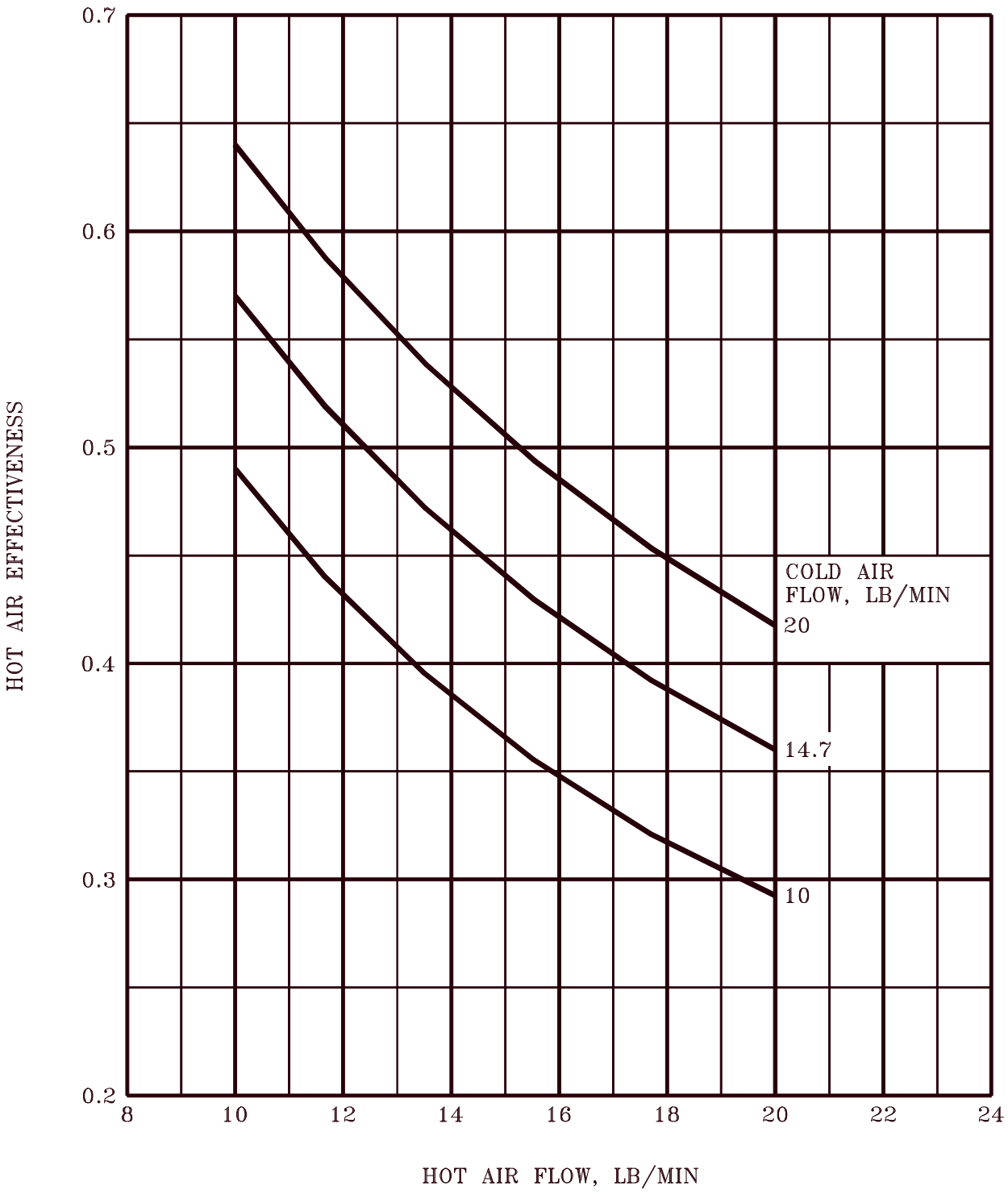


FIGURE 1. Required condenser performance.

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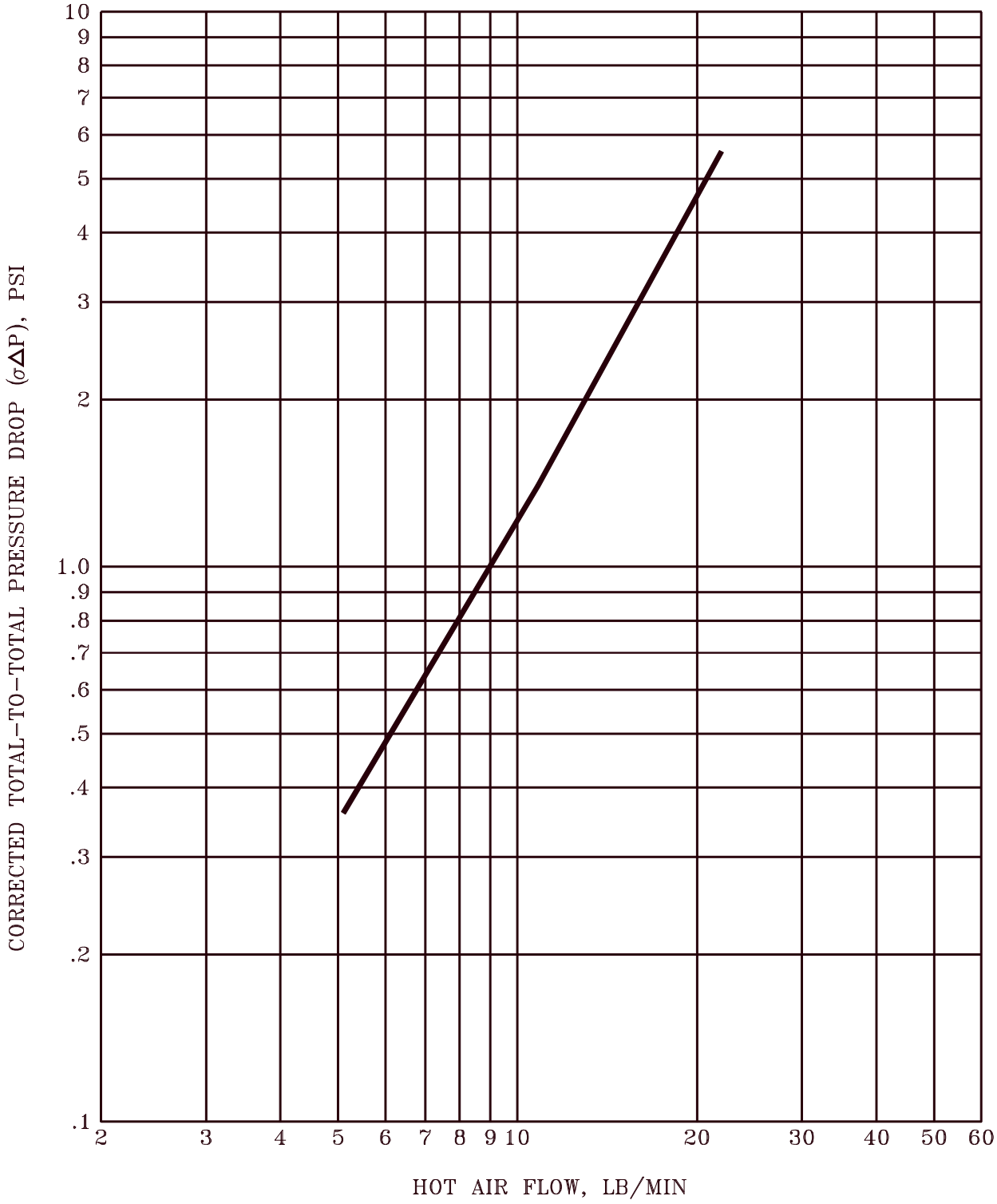


FIGURE 2. Allowable condenser hot air pressure loss.

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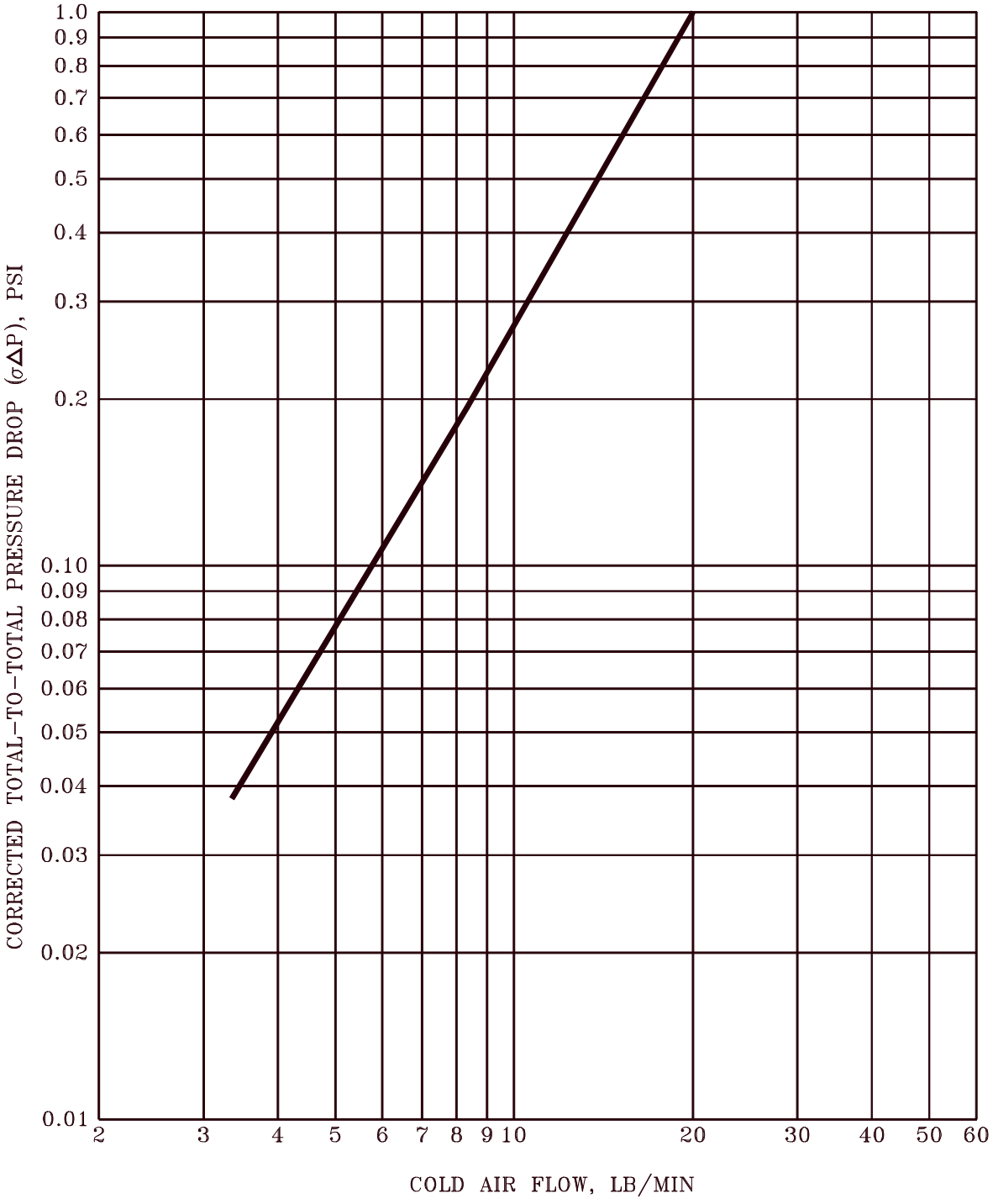


FIGURE 2A. Allowable condenser cold air pressure loss.

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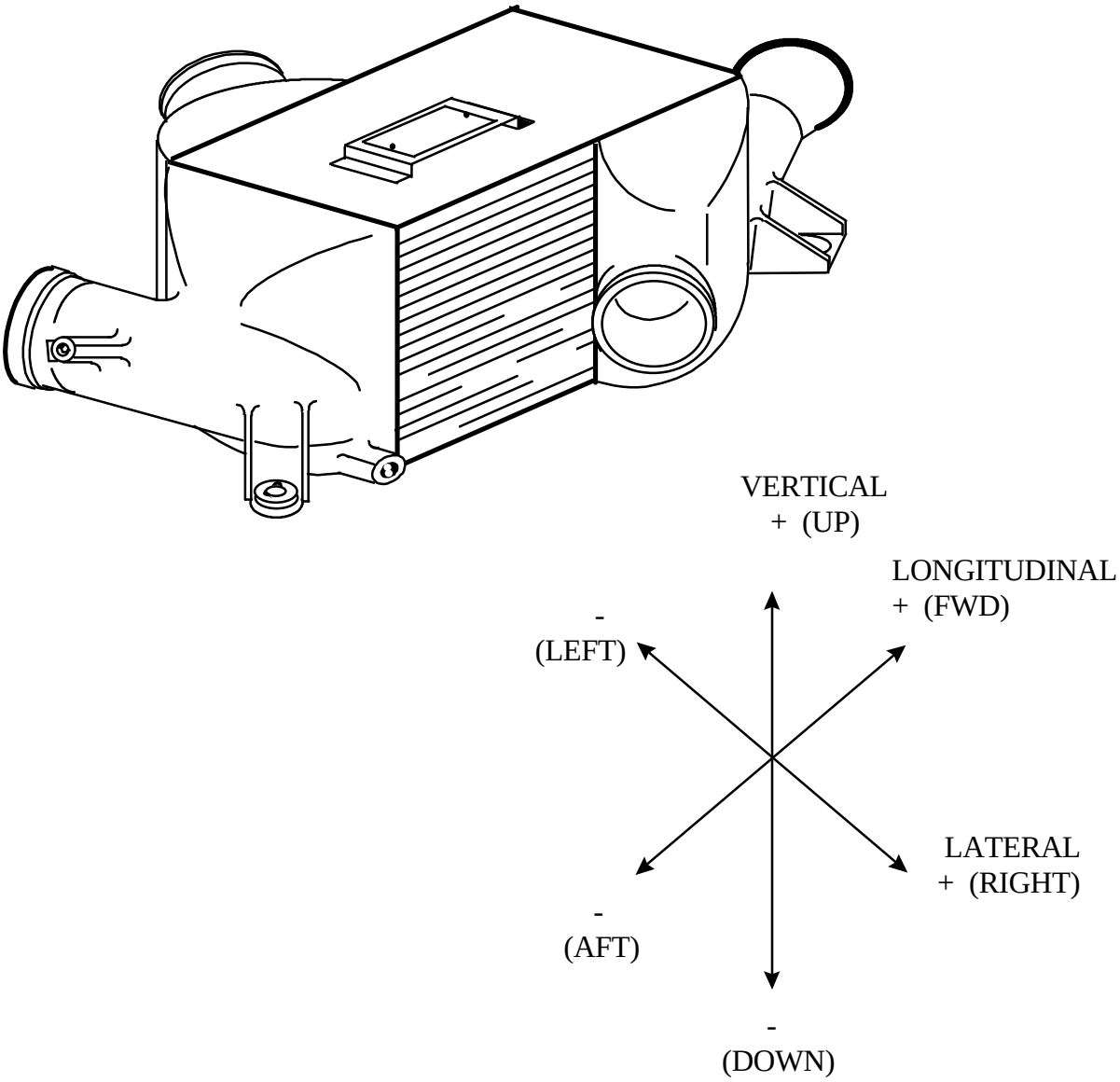


FIGURE 3. Axes orientation.

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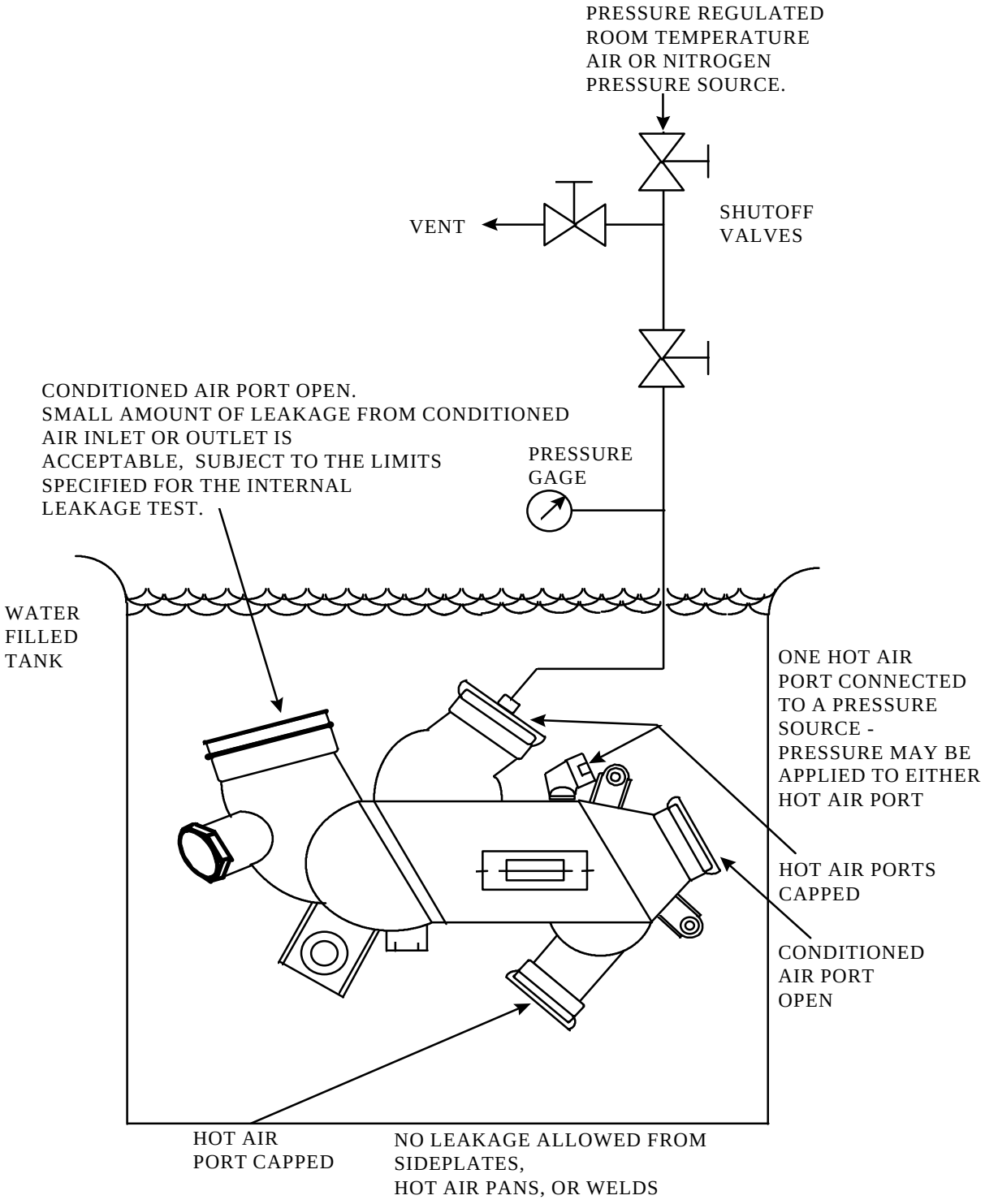


FIGURE 4. Leakage test setup.

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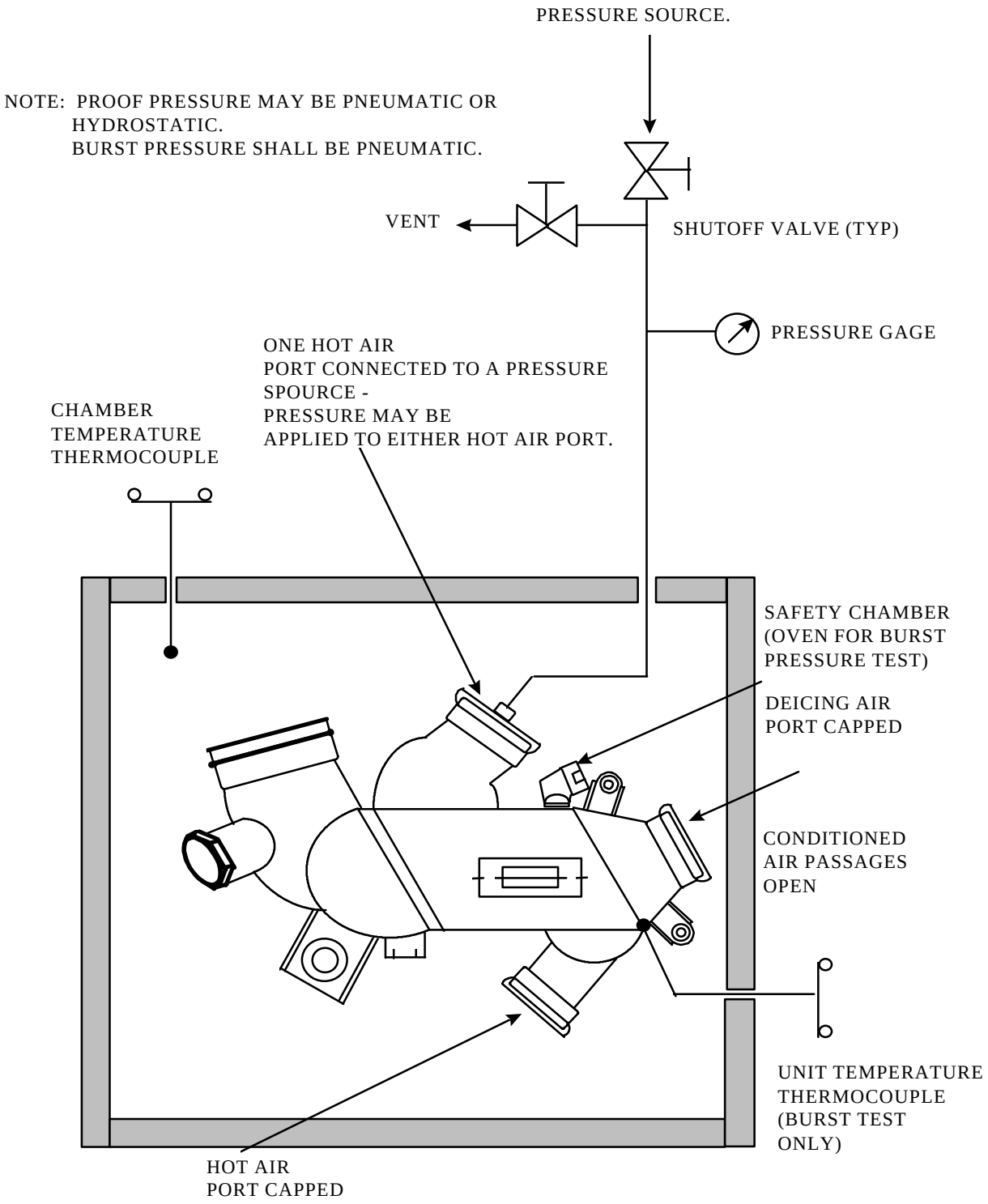


FIGURE 5. Proof and burst pressure test setup.

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